

The Human Respiratory System

An informative passage describing how the body absorbs oxygen and releases carbon dioxide.

The respiratory system is responsible for supplying oxygen to the body and removing carbon dioxide. Humans breathe through the nose, where tiny hair-like structures and mucus help trap dust particles and harmful microorganisms from the air. The air then passes through the trachea and enters the lungs through two branches called bronchi.

Inside the lungs are millions of tiny air sacs known as alveoli. These structures are surrounded by thin blood vessels called capillaries. Oxygen from the inhaled air passes through the walls of the alveoli into the blood, while carbon dioxide from the blood moves into the alveoli to be exhaled. This exchange of gases is essential for producing energy in the body.

Breathing is controlled by the diaphragm and rib muscles. When the diaphragm contracts, the chest cavity expands and air enters the lungs. When it relaxes, the lungs contract and air is pushed out. Maintaining healthy lungs requires regular exercise, clean surroundings, and avoiding harmful habits such as smoking. A healthy respiratory system helps the body function efficiently and supports overall well-being.